

Azure DevOps Service Connection Setup

Service principal and tenant ID field mapping

Use this guide to configure an Azure Resource Manager service connection for Azure US Government using a service principal and client secret. Create the Azure identity, assign access, map the values into Azure DevOps, and verify the connection.

Overview of required values

The service connection uses five Azure values and a name chosen in Azure DevOps.

Value	Azure source	Azure DevOps field
Subscription ID	Subscriptions	Subscription Id
Subscription name	Subscriptions	Subscription Name
Application client ID	App registrations > Overview	Service Principal Id
Client secret value	App registrations > Certificates & secrets	Service Principal Key
Directory tenant ID	App registrations > Overview	Tenant Id

Create the service principal

1 Register the application

Open <https://portal.azure.us> and go to Azure Active Directory (Microsoft Entra ID) > App registrations > New registration.

Field	Value
Name	{project}-{env}-deploy-appreg
Supported account types	Single tenant
Redirect URI	Leave blank

Example application name: myproject-dev-deploy-appreg.

2 Record the application and tenant IDs

After creating the app registration, open its Overview page and record these values.

Azure field	Use
Application (client) ID	Enter as Service Principal Id in Azure DevOps.
Directory (tenant) ID	Enter as Tenant Id in Azure DevOps.
Object ID	For RBAC, use the service principal object ID from Enterprise applications; it is distinct from the app registration object ID.

3 Create a client secret

In the app registration, select Certificates & secrets > New client secret.

Field	Value
Description	ADO service connection
Expires	Choose an expiry allowed by your organization. The example in the guide uses 24 months.

Copy the secret VALUE immediately. It is shown only once. Enter this value as the Service Principal Key in Azure DevOps. The Secret ID is not the credential.

4 Assign an RBAC role

Go to Subscriptions > your subscription > Access control (IAM) > Add role assignment.

Field	Value
Role	Contributor, or a suitable custom role
Assign access to	User, group, or service principal
Select	Search for the application registration name

Assign access at the scope required for deployment. The source example uses the subscription scope.

Organization specific requirement in the source guide: add the service principal to ECS CC All SP Reader Group through a GSEP request for gallery access and shared resources. Confirm this requirement with your internal team.

Create the Azure DevOps service connection

5 Enter the connection details

Open Azure DevOps > Project Settings > Service Connections > New service connection. Select Azure Resource Manager and the manual service principal option using a client secret.

Form field	What to enter
Environment	Azure US Government
Subscription Id	Your Azure subscription ID
Subscription Name	Your Azure subscription name
Service Principal Id	Application (client) ID
Service Principal Key	Client secret VALUE
Tenant Id	Directory (tenant) ID
Service Connection Name	{PROJECT}_AZ_DEPLOY_{ENV}
Pipeline access	The source example grants access to all pipelines. Use your organization's required pipeline authorization settings.

Common confusions

Mistake or question	Correct mapping or action
Object ID entered as Service Principal Id	Use the Application (client) ID.
Secret ID entered as the key	Use the client secret Value.
Subscription tenant versus app tenant	These match when the subscription and application belong to the same tenant.
Commercial Azure selected for a Gov workload	Select Azure US Government.
No RBAC access for the service principal	Assign the deployment permissions at the required scope.

Verify the connection

6 Verify in Azure DevOps

After filling in the fields, click Verify. If verification succeeds, save the connection.

If verification fails, check the secret value, secret expiry, RBAC assignment, selected Azure environment, and tenant ID. A successful login alone does not prove that the identity has all permissions needed by the pipeline.

Verify with Azure CLI

These commands verify login and subscription access. They do not create an Azure DevOps service connection. Replace every placeholder before running.

```
# Login as the service principal
az cloud set --name AzureUSGovernment
az login --service-principal \
  --username "<Application-client-ID>" \
  --password "<Client-secret-value>" \
  --tenant "<Directory-tenant-ID>"

# Verify subscription access
az account show
az group list --output table
```

Naming convention examples

Property	Pattern or example
Service connection	{PROJECT}_AZ_DEPLOY_{ENV} MYPROJECT_AZ_DEPLOY_PROD
App registration	{project}-{env}-deploy-appreg myproject-prod-deploy-appreg
Connection type	Azure Resource Manager using a service principal secret

Quick reference

Application client ID → Service Principal Id
Directory tenant ID → Tenant Id
Client secret VALUE → Service Principal Key
Subscription ID → Subscription Id
Subscription name → Subscription Name
Azure US Government → Environment

Keep client secrets secure and never commit them to source control.